

Network Security Concepts — Guided Notes

Unit 4.1.1 | CompTIA Network+ N10-009 Obj. 4.1 | N10-008 4.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The three pillars of the CIA Triad are _____, _____, and _____. Briefly define each in one sentence.

Question 2

A _____ is someone with authorized network access who misuses it, while a _____ is an outside attacker with no prior access. Which poses the greater risk in most enterprise environments, and why?

Question 3

Explain the difference between a vulnerability, a threat, and a risk. Give a one-sentence example of each in a network context.

Question 4

Zero-day vulnerabilities are dangerous because they have _____ available from the vendor when attackers first exploit them. The primary defense while a patch is unavailable is called _____.

Question 5

Defense in depth means using _____ layers of security controls so that failure of any single control does not result in a complete breach. Name three different layers this strategy might include.

Question 6

The principle of least privilege states that users and systems should have access to only what they _____ to perform their job. How does this limit the damage from a compromised account?

Question 7

Network segmentation limits _____ movement by placing critical systems in separate VLANs or security zones. What firewall behavior enforces the boundary between zones?

Question 8 — Real World Application

REAL WORLD: A school's student VLAN can reach the staff grading portal because no ACL separates them. Which CIA Triad principle is most directly violated, and what single technical control would fix it?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-1-1-network-security-concepts/4-1-1-network-security-concepts-answer-key.pdf
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